

● LEVEL 1 — Basics & Array Thinking (Q1–Q8)

Goal: Become fluent with array creation, indexing, and simple operations.

1. A sensor records temperature every hour for 24 hours.
Create a NumPy array of 24 values between **15 and 35** (inclusive).
 2. Given an array of daily sales:
[1200, 1500, 1100, 1800, 1700]
 - Find **maximum**, **minimum**, and **average** sales.
 3. A student has marks in 6 subjects.
Create an array and:
 - Add **5 grace marks** to each subject.
 4. You have an array of ages of people in a survey.
Extract all ages **greater than 18**.
 5. Create a **2×3 matrix** representing sales of 2 products over 3 days.
Print:
 - Row-wise sum
 - Column-wise sum
 6. Reverse a NumPy array **without using loops**.
 7. Given [10, 20, 30, 40, 50], replace:
 - values **> 25** with 1
 - values **≤ 25** with 0
 8. Check whether an array contains **any negative values**.
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● LEVEL 2 — Indexing, Broadcasting, Shape Logic (Q9–Q16)

Goal: Learn slicing, broadcasting, reshaping, and boolean logic.

9. A company stores monthly profits for 3 years in a **1D array (36 values)**.
Extract profits of:
 - Year 2 only
 - Last 6 months
10. Given a 5×5 matrix of employee performance scores:
 - Extract the **middle 3×3 sub-matrix**

11. Normalize an array of exam scores to range **0–1** using NumPy.

12. Given two arrays:

- product prices
 - discount percentages
- Compute **final prices** using broadcasting.

13. Replace all **odd numbers** in an array with -1.

14. Convert a **1D array of 12 values** into:

- 3×4 matrix
- 4×3 matrix

15. Count how many values in an array are:

- above mean
- below mean

16. Given daily step counts of 30 days:

- Find the **longest streak** of days with steps > 8000.

● LEVEL 3 — Statistics, Missing Values, Real-World Patterns (Q17–Q23)

Goal: Think like a data analyst.

17. A dataset contains NaN values in salary data.

- Replace missing salaries with **mean salary**.

18. Given a dataset of product ratings (1–5):

- Find **most frequent rating**
- Find **standard deviation**

19. Detect **outliers** in an array using:

- $\text{mean} \pm 2 \times \text{std deviation}$

20. You have rainfall data for 12 months:

- Identify **dry months** (rainfall < 50)
- Count them

21. Given marks of students:

- Assign grades:
 - $\geq 90 \rightarrow A$
 - $\geq 75 \rightarrow B$
 - $\geq 60 \rightarrow C$
 - else $\rightarrow D$*(Use NumPy only, no loops)*

22. Compare two exam result arrays and find:

- how many students **improved**
- how many **declined**

23. Given stock prices of 100 days:

- Compute **daily returns (%)**
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● LEVEL 4 — Data Science Mindset (Q24–Q30)

Goal: Analytics + performance + thinking beyond syntax.

24. Simulate **1000 random salaries** (mean=50k, std=10k).

- Find percentage of people earning **> 70k**

25. Given user session durations:

- Bin them into:
 - short (<5 min)
 - medium (5–30 min)
 - long (>30 min)

26. A machine generates sensor data every second (10,000 points).

- Smooth the signal using **moving average**.

27. Given customer purchase amounts:

- Identify **top 10% spenders**.

28. You have a matrix of website traffic:

- rows $\rightarrow$ days

- columns $\rightarrow$ hours

Find:

- busiest hour overall

29. Compare execution time of:

- Python loop vs NumPy vectorized operation
(*Conceptual + implementation*)

30. Given features of a dataset:

- Perform **z-score normalization**
- Verify mean ≈ 0 and std ≈ 1